



STEADYWIN[®] Electric Drive MIT Interface Protocol Overview

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1 MIT CAN Protocol

The open CAN protocol for the **SteadyWin®** GIM electric drive is part of the MIT open-source protocol.

1.1 Host Computer→ Electric Drive

The data sent from the host computer to the drive is 8 bytes long, in a standard data frame.

➤ Retrieve/Clear Exceptions

Frame Bit	BYTE0	BYTE1	BYTE2	BYTE3	BYTE4	BYTE5	BYTE6	BYTE7
Data	0xFF	0xFF	0xFF	0xFF	0xFF	0xFF	F_CMD	0xFB

If the F_CMD byte is 0xFF, it indicates that the current exception is cleared; if it is any other value, the exception code will be returned in BYTE1 of the response.

➤ Start motor

Frame Bit	BYTE0	BYTE1	BYTE2	BYTE3	BYTE4	BYTE5	BYTE6	BYTE7
Data	0xFF	0xFF	0xFF	0xFF	0xFF	0xFF	0xFF	0xFC

➤ Stop the motor

Frame position	BYTE0	BYTE1	BYTE2	BYTE3	BYTE4	BYTE5	BYTE6	BYTE7
Data	0xFF	0xFF	0xFF	0xFF	0xFF	0xFF	0xFF	0xFD

➤ Set zero point

Frame position	BYTE0	BYTE1	BYTE2	BYTE3	BYTE4	BYTE5	BYTE6	BYTE7
Data	0xFF	0xFF	0xFF	0xFF	0xFF	0xFF	0xFF	0xFE

➤ Dynamic Control

Number of CAN Data frame bits	Meaning	Description
BYTE0	Position: Total of 16 bits; BYTE0 is the upper 8 bits, BYTE1 is the lower 8 bits Multi-turn position of the output shaft, in radians (RAD)	Actual type is double; must be converted to 16 bits of type int; the conversion process is: $\text{pos_int} = (\text{pos_double} + 12.5) * 65535 / 25$
BYTE1		

	Speed: 12 bits total, with BYTE2 comprising the upper 8 bits, BYTE3[7-4] (the lower 4 bits) are the lower 4 bits. Represents the angular velocity of the output shaft, in units of RAD/s KP value: 12 bits in total, with BYTE3[3-0] (the lower 4 bits) are the high 4 bits, and BYTE4 is the low 8 bits.	The actual speed is of type double and needs to be converted to a 12-bit int, as follows: $vel_int = (vel_double + 65) * 4095 / 130$ The KP value is actually of type double and needs to be converted to a 12-bit int type; the conversion process is: $kp_int = kp_double * 4095 / 500$
BYTE2		
BYTE3		
BYTE4	KD value: 12 bits total, with BYTE5 representing the upper 8 bits , with BYTE6[7-4] (the upper 4 bits) representing the lower 4 bits.	The KD value is actually of type double and needs to be converted to 12 bits int type; the conversion process is: $kd_int = kd_double * 4095 / 5$
BYTE5		
BYTE6	Torque: Total of 12 bits; BYTE6[3-0] (low 4	The actual torque is of type double and needs to be converted to 12 bits
BYTE7	BYTE6 contains the upper 4 bits, and BYTE7 contains the lower 8 bits. The unit is N-m.	int type; the conversion process is as follows: $t_int = (t_double + 225 * torque\ constant * reduction\ ratio) * 4095 / (450 * torque\ constant * reduction\ ratio)$ The unit of the torque constant is N-m/A

1.2 Electric Drive→ Host Computer

The data sent by the electric drive to the host computer is 6 bytes long, in a standard frame.

➤ Retrieving/Clearing Fault Responses

When retrieving an exception command, BYTE0 is the CAN ID, BYTE1 is the exception number, and the other bytes are meaningless. The following table lists the general exception numbers:

FaultNo	Description
0x00	No fault
0x01	FoC frequency too high
0x02	Overvoltage
0x04	Undervoltage
0x08	Overtemperature
0x10	Startup Failure

0x40	Overcurrent
0x80	Software exception

When responding to clear an exception, BYTE0 is the CAN ID; the other bytes are meaningless.

➤ Other command responses

CAN number	Meaning	Description
Data		
Frame		
Bits		
BYTE0	CAN ID	Electric Drive CAN ID
BYTE1	<u>Position</u> : 16 bits total, with BYTE1 comprising the upper 8 bits,	<u>The actual position</u> is of type double and must be converted from a 16-bit int. The conversion process is:
BYTE2	BYTE2 is the lower 8 bits	$pos_double = pos_int * 25 / 65535 - 12.5$
	Multi-turn position of the output shaft, in radians (RAD)	
BYTE3	<u>Speed</u> : 12 bits total, with BYTE3 representing the upper 8 bits,	<u>The actual velocity</u> is of type double and needs to be converted from a 12-bit int. The conversion process is as follows:
BYTE4	BYTE4[7-4] (the upper 4 bits) are the lower 4 bits. Represents the angular velocity of the output shaft, in units of RAD/s	$vel_double = vel_int * 130 / 4095 - 65$
BYTE5	<u>Torque</u> : Total of 12 bits; BYTE4[3-0] (the lower 4 bits) contains the upper 4 bits, and BYTE5 contains the lower 8 bits. The unit is N·m.	<u>The actual torque</u> is of type double and must be converted from a 12-bit int. The conversion formula is: $t_double = t_int * (450 * torque\ constant * reduction\ ratio) / 4095 - 225 * torque\ constant * reduction\ ratio$ The unit of the torque constant is N·m/A